Infobits

i) Isotype

Isotypic determinants are constant region determinants that collectively define each heavy chain class and subclass and each light chain type and subtype within a species. Each isotype is encoded by a separate constant region gene and all members of a species carry the same constant region genes. Within a species, each normal individual will express all isotypes in the serum. Different species inherit different constant region genes and therefore express different isotypes. Therefore, when an antibody from one species is injected into another species, the isotypic determinant will be recognized as foreign, inducing an antibody response to the isotypic determinants on the foreign antibody.

ii) Allotype

Although all members of a species inherit the same set of isotype genes, multiple alleles exist for some of the genes. These alleles encode subtle aminoacid differences, called **allotypicdeterminnats**that occur in some.

The unique amino acid sequence of the VH and VL domains of a given antibody can function not only as an antigenic binding site but also as a set of antigenic determinants. Therefore, the idiotypic determinants are generated by the conformation of the heavy and light chain variable regions. Each individual determinant is called an idiotope and the sum of the individual idiotopes is the idiotype. Anti-idiotype antibody is produced by injecting antibodies that have minimal variation in their isotypes and allotypes, so that the idiotypic difference can be recognized.

13.6.4 Antigenic Determinants on Immunoglobulins

Since antibodies are glycoproteins, they can themselves function as potent immunogens to induce an antibody response. Such anti-Ig antibodies are powerful tools for the study of B cell development and humoral immune response. The antigenic determinants or epitopes, on immunoglobulin molecules fall into three major categories: isotypic, allotypic and idiotypicdeteminants, which are located in characteristic portions of the molecule.

13.7 Antigen – Antibody Reactions

Antigen and antibody combine with each other specifically and in an observable manner. The exquisite specificity of antigen-antibody interactions has led to the development of a variety of immunological assays. These assays can be used to detect the presence of either antibody or antigen. These assays are also helpful in diagnosing diseases, monitoring epidemiological surveys and identifying molecules of biological or medical interest. Antigen-antibody reactions in vitro are known as serological reactions.

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